Chapter 19, Problem 10P

Problem

Construct a two-port that realizes each of the following *z* parameters.

(a) $[\mathbf{z}] = \begin{bmatrix} 25 & 20 \\ 5 & 10 \end{bmatrix} \Omega$

(b) $[\mathbf{z}] = \begin{bmatrix} 1 + \frac{3}{s} & \frac{1}{s} \\ \frac{1}{s} & 2s + \frac{1}{s} \end{bmatrix} \Omega$

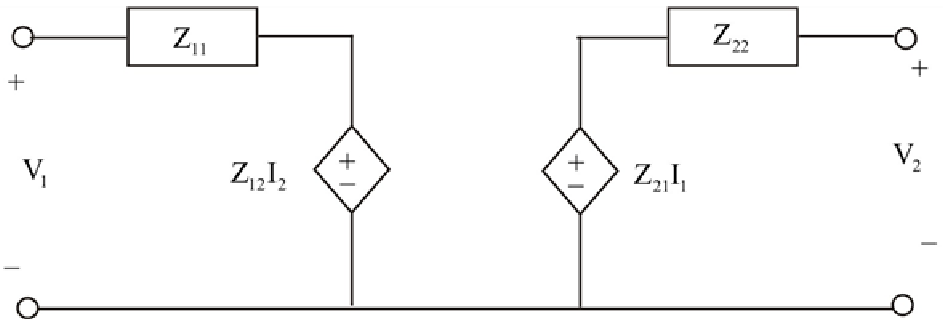
Step-by-step solution

Step 1 of 4

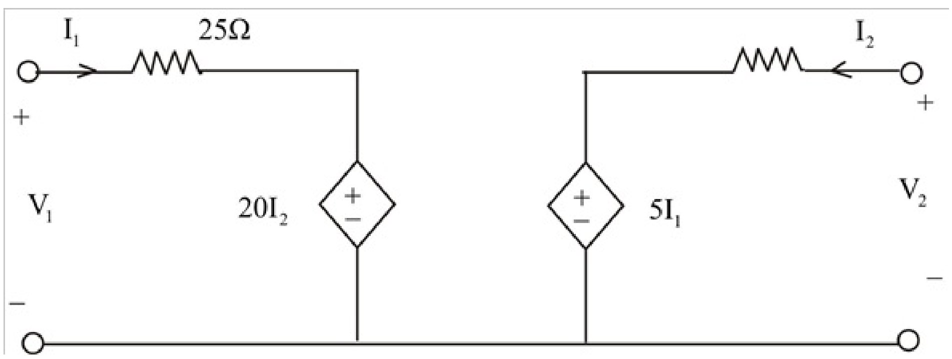
(a) $(Z) = \begin{bmatrix} 25 & 20 \\ 5 & 10 \end{bmatrix} \Omega$

$V_1 = Z_{11}I_1 + Z_{12}I_2$

$V_1 = Z_{21}I_1 + Z_{22}I_2$



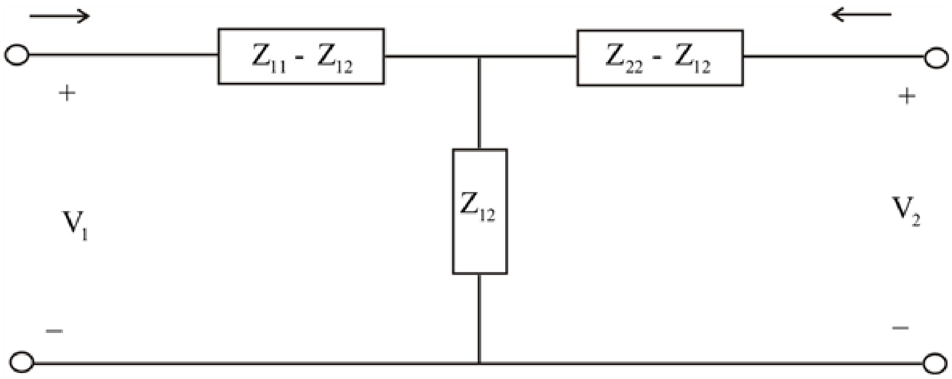
Step 2 of 4



Step 3 of 4

(b) $$(Z) = \begin{bmatrix} \frac{1}{3s} & \frac{1}{s} \\ \frac{1}{s} & 2s+\frac{1}{s} \end{bmatrix} \Omega$$

Since $Z_{12} = Z_{21}$ the two port network is symmetrican



Step 4 of 4

$$Z_{11}\text{-}Z_{12} = 1+\frac{3}{s} - \frac{1}{s} = 1+\frac{2}{s}$$

$$Z_{22} - Z_{12} = 2s-\frac{1}{s}-\frac{1}{s} = 2s$$